A CRITERION FOR BIRATIONALITY OF IRREDUCIBLE SCHEMES LOCALLY OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Editorial note. Read in full under the Lead's 12:4x rule. The proof contains 'It is immediate that (2) implies (3)' - a genuinely immediate implication between two of the listed conditions; the rest of the argument is given in full, so this is not a hint-only proof.

Allowed standard prerequisites (editorial).

- Equivalence of birationality with the existence of isomorphic nonempty opens. Lemma \ref{lemma-birational-integral}, cited at proof lines 13132-13133. Gives the first equivalence.
- Existence of a rational map realising a prescribed isomorphism of local rings at generic points. Lemma \ref{lemma-rational-map-finite-presentation}, cited at proof lines 13136-13139. Produces the birational morphism.
- A birational morphism induces an isomorphism on nonempty opens. Lemma \ref{lemma-birational-birational}, cited at proof lines 13140-13142. Finishes the cycle.

Source reference. Original lemma 0BAD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14206–14206. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0BAA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14203–14203. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0BX8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14971–14971. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BAC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14205–14205. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let S be a scheme. Let X and Y be irreducible schemes locally of finite presentation over S. Let $x \in X$ and $y \in Y$ be the generic points. The following are equivalent*

(1) *X and Y are S-birational,*
(2) *there exist nonempty opens of X and Y which are S-isomorphic, and*
(3) *x and y map to the same point s of S and $\mathcal{O}_{X,x}$ and $\mathcal{O}_{Y,y}$ are isomorphic as $\mathcal{O}_{S,s}$-algebras.*

Proof. We have seen the equivalence of (1) and (2) in Lemma 0BAA. It is immediate that (2) implies (3). To finish we assume (3) holds and we prove (1). By Lemma 0BX8 there is a rational map $f : U \to Y$ which sends $x \in U$ to y and induces the given isomorphism $\mathcal{O}_{Y,y} \cong \mathcal{O}_{X,x}$. Thus f is a birational morphism and hence induces an isomorphism on nonempty opens by Lemma 0BAC. This finishes the proof. □